Fair Data Management in a Cloud Environment

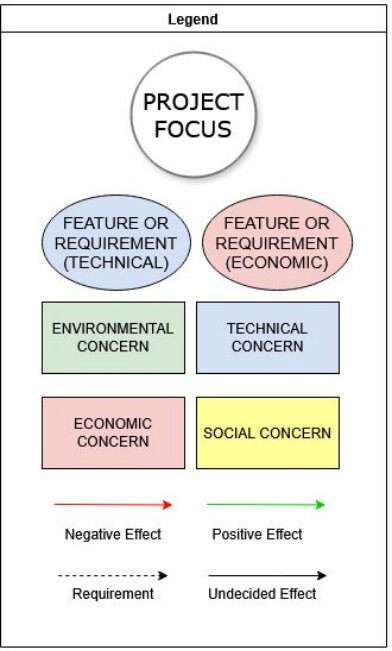
D4F Assignment*

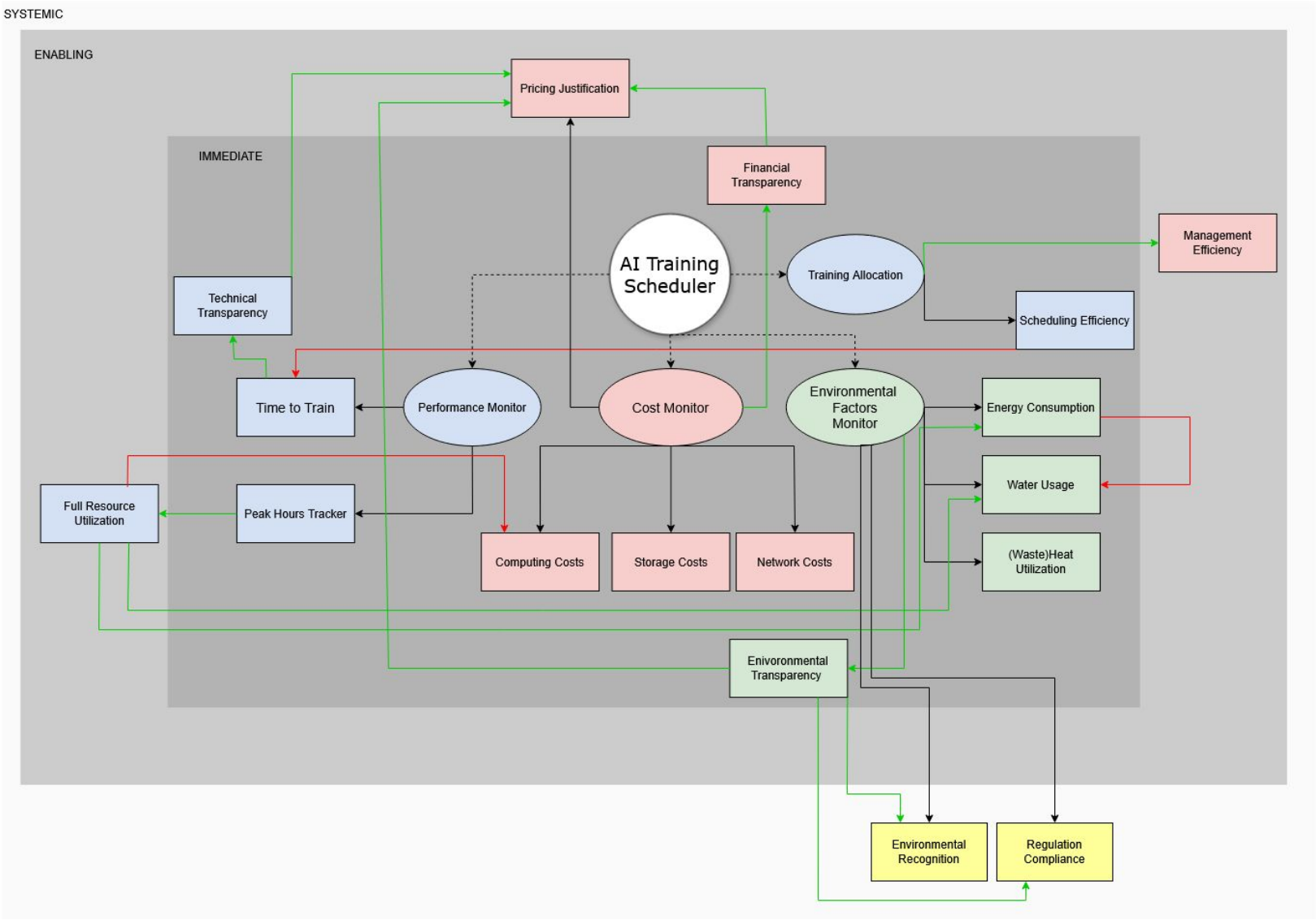
Use case: FairTrain- Sustainable AI Scheduling

- ## Scenario
  Alice, a machine learning engineer at a tech startup, manages AI training jobs on the cloud to optimise the performance of various services offered by the company. The company has long set out goals to reduce its carbon footprint, but Alice lacks visibility into the environmental impact of different data centers(DCs) available for assignment. Training jobs are currently scheduled based on cost and speed, not sustainability. Alice needs a solution that minimises emissions, while maintaining cost-efficiency.
- ## Expectations
  Alice using the cloud for a machine learning job should be able to select conditions that align with her company's sustainability minded practices. She should be able to trust that the system will take those conditions in consideration when assigning her job to a data center.
- ## Proposed System and its main tasks
  A cloud AI training scheduler systems that:
  1. monitors environmental impact, costs, and performance of individual DCs.
  2. optimise the scheduling of cloud training jobs by prioritising data centres with lower environmental impact and shifting workloads to off-peak hours, all while keeping costs reasonable.
- ## Environment
  Centralised, cloud-based AI model training environment.
- ## Current state and Vision
  In today's world, the environmental impacts of DCs are rarely measured or reported at a regulatory level [1], let alone made visible to users like Alice.
  Our vision is to create transparent, environmentally-aware AI training that balances sustainability, cost, and performance.

[1] D. Mytton and M. Ashtine, 'Sources of data center energy estimates: A comprehensive review', *Joule*, vol. 6, no. 9, pp. 2032–2056, Sep. 2022, doi: 10.1016/j.joule.2022.07.011.

Definition of Fairness: Decision Map





Definition of Fairness: Decision Map (cont.)

The system can be broken down into four main components: Environmental Factors Monitor, Cost Monitor, Performance Monitor, and finally, Training Allocation.

All three monitors collects data from individual data centres(DCs) that are part of the cloud training environment.

- The **Environmental Factors Monitor** collects individual DCs' data on their impact on the environment, these data are roughly grouped under four environmental concerns: **Energy Consumption** concerns data such as total energy consumption, IT equipment energy consumption, and etc. **Water Usage** concerns the total water usage of individual DCs. (**Waste)Heat Utilisation** concerns data about a DC's total amount of waste heat and utilised waste heat. All together, these data collected helps calculate metrics such as Power Usage Effectiveness, Carbon Intensity, and etc, about each DC(These metrics are explained in greater detail in the KPI canvas). These data are updated to the system daily, and enables **Environmental Transparency**.
- The **Cost Monitor** and **Performance Monitor** collect data to estimate the cost and performance of individual data centers (DCs) for each training job scheduled by the system. This data is calculated each time a job enters the system for scheduling.The Cost Monitor tracks the financial cost incurred by each job at a data center, including **Computing**, **Storage**, and **Network costs**. These three components are combined to calculate the total cost per job. The Performance Monitor records the estimated **time-to-train** for each job and monitors whether the data center is operating during peak or off-peak hours(**Peak Hours Tracker**). Data from these two monitors enables **Financial Transparency** and **Technical Transparency**.

Finally, the **Training Allocation** access information collected by the three monitors, scheduling jobs in the cloud system. This enables **Scheduling Efficiency,** which uses up to date information to assess which DC is the most suitable target for assignment, which in turn creates **Management Efficiency.** The entire process is illustrated in detail in the fairness view by the end of the slides.

We considered **Environmental, Financial,** and **Technical Transparency**, as well as **Scheduling Efficiency** immediate impacts of the system, because they are addressed directly within the system, and their effects are immediately visible ones the system starts operating. **Pricing Justification**, **Management Efficiency**, and **Full Resource Utilisation**, are enabled by the immediate impacts, and only rises from use over time. On a macro level, the system have an impact for Environmental Recognition for DCs, as well as being able to reason about their **Regulation compliance**, thus, these two are considered systemic impacts[1].

[1]P. Lago, 'Architecture Design Decision Maps for Software Sustainability', in *2019 IEEE/ACM 41st International Conference on Software Engineering: Software Engineering in Society (ICSE-SEIS)*, Montreal, QC, Canada: IEEE, May 2019, pp. 61–64. doi: 10.1109/icse-seis.2019.00015.

Definition of Fairness: SQ model

QUALITY ATTRIBUTE (QA)		DEFINITION	REFERENCE	TECHNICAL	ENVIRONMENTAL	ECONOMIC	SOCIAL
Transparency	Enviromental Transparency	the degree to which a system discloses information regarding its effect on the environment			X		
	Financial Transparency	the degree to which a system discloses information regarding pricing practices				X	
	Technical Transparency	the degree to which a system discloses information regarding technical details		X			
Efficiency	Management Efficiency	manangement resources expended in relation to the accuracy and completeness with which users achieve goals	[ISO/IEC 25010:2011]			X	
	Scheduling Efficiency	the time it takes to efficiently schedule a job to the appropriate data center depending on resource metrics		X			
	Full Resource Utilization	degree to which the amounts and types of resources used by a system, when performing its functions, meet requirements	[ISO/IEC 25010:2011]	X			
Regulation Compliance		the degree to which a system allows to draw conclusions about how well the software adheres to application related regulations in laws	adapted from [Condori-Fernandez et al. (2020)]				X
Pricing Justification		the degree to which a systems enables reasoning behind pricing practices in regards to other factors, such as environmental-friendliness				X	
Environmental Recognition		the degree to which a system facilitates recognation of its components for being environmental-friendly					X
Costs	Computing Costs	the cost of processing power required over a period of time				X	
	Storage Costs	the cost of storing data on physical hardware				X	
	Network Costs	the cost of network traffic				X	
Water Usage		the amount of water used for hardware cooling purposes			X		
Energy Consumption		the total energy consumption of a data center			X		
Waste Heat Utilization		the degree of waste heat generated being re-used for other beneficial purposes			X		
Time to Train		the average time it takes for AI training job completion		X			
Peak Hours Tracker		a set of time periods per data center that the data center experiences close to max capacity		X			

Definition of Fairness: SQ model (cont.)

In the SQ model, all the concerns from our decision map are grouped and defined as quality attributes(QAs), as well as mapped for their sustainability dimensions.
The QAs can roughly be divided into two groups, group 1 are filled in light grey in the SQ model, while group 2 are Filled in darker grey.

<u>Group 1:</u>

Group 1 are typical QAs and are defined by the degree to which the system archives them.
We have mapped **Environmental, Financial, and Technical Transparencies** in the Environmental, Economical, and Technical dimensions respectively, which is self-explanatory. We have also grouped these three transparencies under one "Transparency" group. They are all important to the fairness of our system, as they discloses information about DCs in the system that are not typically visible to the users.
"Efficiency" is another group of QAs in Group one. **Management Efficiency**, which potentially reduces overhead and saves budget, is in the economic dimension. **Scheduling Efficiency** and **Full Resource Utilisation**, as explained in slide 4, are technical QAs that concerns the performance of the system, thus they are mapped in the technical dimension. The Efficiency group of QAs are important to fairness of our system, because if they are not balanced with reducing environmental impact, users won't potentially be incentivised to take part in the system, reducing its potential positive impact on the environment.
Regulatory Compliance is mapped in the social dimension because it impacts the social functioning and image of the entity concerned. Being able to reason about how a DC compliance with regulation is important for the system's fairness. Same with **Environmental Recognition**.
Finally, **Pricing Justification** is again important for the fairness of the system, because DCs who are more environmental friendly can argue that their higher pricing is fair, and vice-versa.

<u>Group 2:</u>

QAs in group 2 are more specific to our system's use case, as they measures individual DCs' parameters, and they are mapped according to the parameters they measure. Nonetheless, the QAs in Group 2 are extremely important to the system, as without them, there wouldn't be the data availability for the scheduler to make fair decisions, balancing the parameters measured by these QAs.

Fairness perspective

	DESCRIPTION
DESIRED QUALITY	The ability of the system to schedule training based on environmental impact while balancing performance and cost, reducing the environmental impact of AI training at large.
CONCERNS	• Environmental transparency • Financial transparency • Technical transparency • Management efficiency • Pricing justification • Environmental recognition • Resource utilisation • Scheduling Efficiency
ARCHITECTURAL TACTIC(S)	**AT-1**: Implement carbon footprint aware scheduling **AT-2**: Select data centers based on adaptive desirability functions[3] **AT-3**: Design runtime process migration to more suitable data centers [4] **AT-4**: Implement monitoring of environment of individual jobs in progress
POSSIBLE KPI(S)	**KPI 1**: Reduced Environmental Impact Score(see KPI canvas in the next slide); **KPI 2**: Job Scheduling Latency (the time it takes for the system to assign a job to a data center after receiving the request); **KPI 3**: Job Reassignment Rate (percentage of jobs re-assigned to a different DC before completion), **KPI 4**: Average Cost Variance[1] (average difference between the system's estimated costs and the actual cost for AI training jobs)
PROBLEMS AND PITFALLS	• Data availability issues caused by reluctance of DCs to disclose environmental, financial, and technical information without legal renforcement, in fear of negative effects on their reputation[2]. • Scheduling conflicts and overload (suboptimal scheduling strategy) • inaccurate performance and costs predictions leading to lack of trust in system.

[1]AccountingTools, "What is a Cost Variance?" AccountingTools, [Online]. Available: https://www.accountingtools.com/articles/what-is-a-cost-variance.html.

[2] D. Mytton and M. Ashtine, 'Sources of data center energy estimates: A comprehensive review', *Joule*, vol. 6, no. 9, pp. 2032–2056, Sep. 2022, doi: 10.1016/j.joule.2022.07.011.

[4] Alourani A, Khalid A, Tahir M, Sardaraz M (2024) Energy efficient virtual machines placement in cloud datacenters using genetic algorithm and adaptive thresholds. PLOS ONE 19(1): e0296399. https://doi.org/10.1371/journal.pone.0296399

[5] M. Imran, M. Ibrahim, M. S. U. Din, M. A. U. Rehman, and B. S. Kim, "Live virtual machine migration: A survey, research challenges, and future directions," *Computers and Electrical Engineering*, vol. 103, p. 108297, 2022, doi: 10.1016/j.compeleceng.2022.108297.

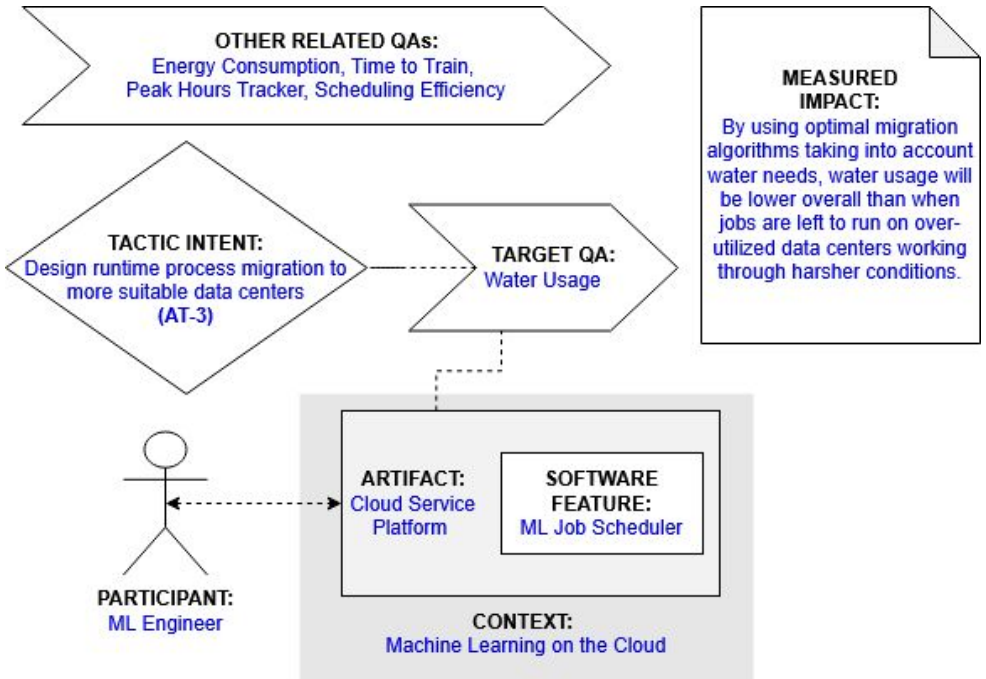
Definition of KPI*

Reduced Environmental Impact Score (REIS)					
Goal	**Critical Success Factor (CSF)**	**Key Performance Indicator (KPI)**		**Metric(s)**	**Measure(s)**
Reduce the overall environmental impact of cloud AI training by dynamically scheduling training job requests to suitable DCs	Have a data-driven assessment of the reduced environmental impact of the system in a given time frame	**REIS =** 25% · ((sum of baseline **PUE**)/(sum of current **PUE**)) + 15% · ((sum of baseline **WUE**)/(sum of current **WUE**)) + 30% · ((sum of baseline **Carbon Intensity**)/(sum of current **Carbon Intensity**)) + 10% · ((sum of baseline **Waste Heat Utilisation Rate**)/(sum of current **Waste Heat Utilisation Rate**) + 20% · ((sum of baseline **Renewable Energy Use Rate**)/(sum of current **Renewable Energy Use Rate**) **Unit: %** note: the weight(%) for each component is arbitrary and can be adjusted if priorities change		**Power Usage Effectiveness (PUE) = (Total Energy Consumption) / (IT Equipment Energy)** **unit: %** [1] **Water Usage Effectiveness(WUE)= (Total Water Usage / (IT Equipment Energy)** **unit: litre/kWh** [1] **Carbon Intensity = (Total CO_2 Emissions) / (Total Energy Consumption)** **unit: kg/kWh** [2] **Waste Heat Utilisation Rate = (Utilised Waste heat) / (Total Waste Heat)** **unit: %** **Renewable Energy Use Rate = (Renewable Energy Consumption) / (Total Energy Consumption)** **unit: %**	**Total water usage** (litres) **Renewable energy consumption** (kWh) **IT Equipment Energy** (kWh) **Total Energy Consumption**(kWh) **Utilised Waste Heat** (kWh) **Total Waste Heat** (kWh) **Total CO_2 Emissions** (kg)
		Target	**Action(s)**		
		if REIS is below 90%, the target is to grow 5% per quarter.	Optimise scheduling strategy according to priorities. For DCs in the system: optimise power management, implement water-saving technologies, invests in renewable energies.		

[1] B. Ristic, K. Madani, and Z. Makuch, 'The Water Footprint of Data Centers', *Sustainability*, vol. 7, no. 8, pp. 11260–11284, Aug. 2015, doi: 10.3390/su70811260.

[2] W. Graus and E. Worrell, 'Methods for calculating CO2 intensity of power generation and consumption: A global perspective', *Energy Policy*, vol. 39, no. 2, pp. 613–627, Feb. 2011, doi: 10.1016/j.enpol.2010.10.034.

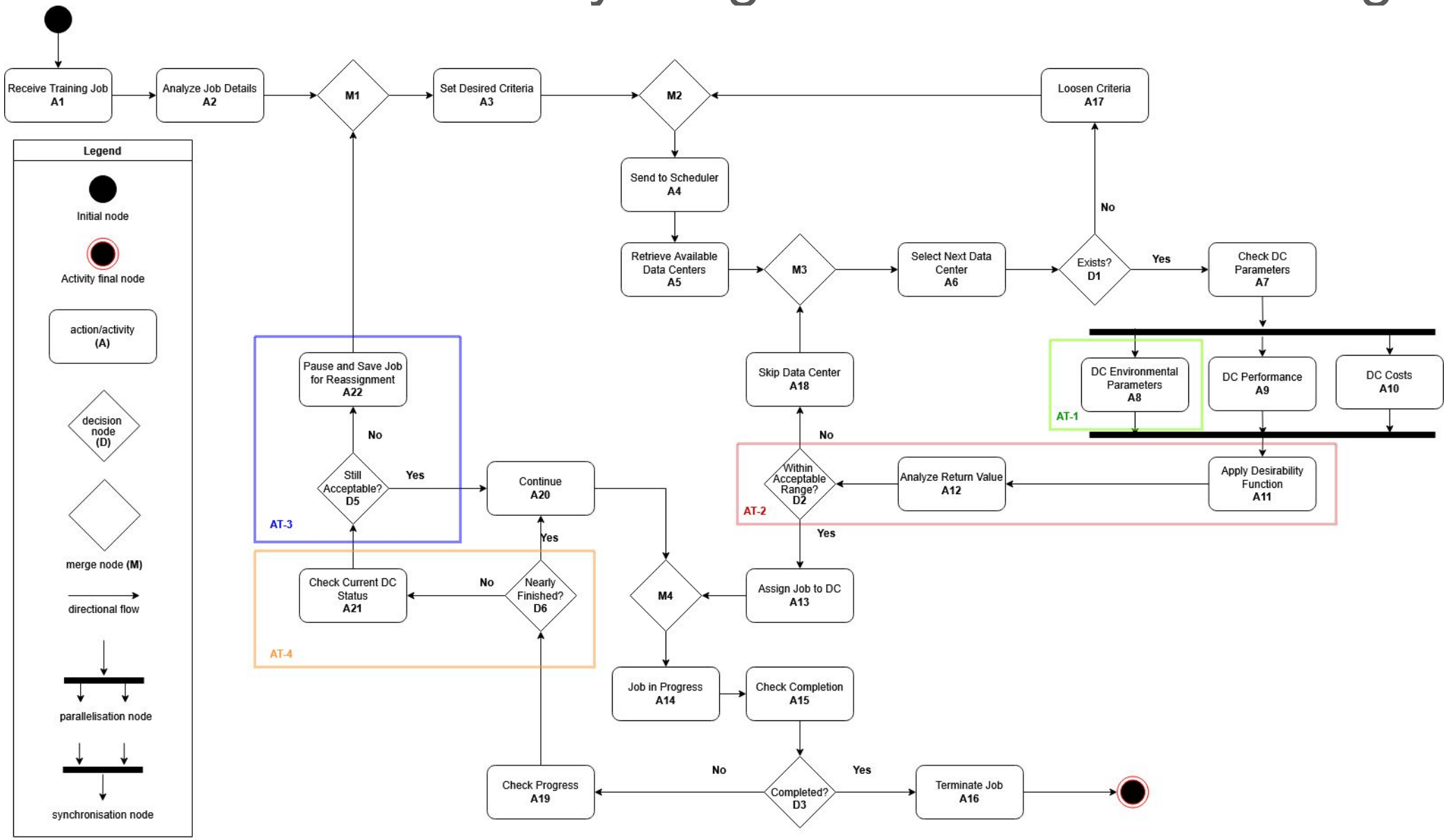
Architectural Tactic(s) for Fairness*



[1] S. S. Panwar, M. M. S. Rauthan, and V. Barthwal, "A systematic review on effective energy utilization management strategies in cloud data centers," *Journal of Cloud Computing: Advances, Systems and Applications*, vol. 11, no. 95, pp. 1–29, 2022, doi: 10.1186/s13677-022-00368-5.
[2] P. Li, J. Yang, M. A. Islam, and S. Ren, "Making AI Less 'Thirsty': Uncovering and Addressing the Secret Water Footprint of AI Models," arXiv preprint arXiv:2304.03271v5, Mar. 2025. [Online]. Available: https://arxiv.org/abs/2304.03271

Title	**AT-3**: Design runtime process migration to more suitable data centers
Description	This refers to moving an ongoing machine learning job on the cloud from one data center to another data center that is currently considered more environmentally friendly and then continue seamlessly, meaning without loss of data, loss of progress or too much downtime. Process migration is possible with machine learning jobs if they are compartmentalized into containers or separate virtual machines using states or checkpoints to pause and move them.
Participant	Machine Learning Engineer
Software Artifact	Cloud Service Platform
Context	Machine Learning on the Cloud
Software Feature	Machine Learning Job Scheduler
Tactic Intent	Decrease job load of data centers during periods of peak energy consumption and water usage needs for cooling due to weather, time of day or above average utilization rate.
Target Quality Attribute	Water Usage
Other Related QAs	Energy Consumption, Scheduling Efficiency, Peak Hours Tracker, Time to Train
Measured Impact	Benefits to energy consumption are apparent when VM migration is implemented to balance load onto underutilized Data Centers by up to 90%[1]. When optimized to take into account factors such as temperature, weather and time of day [2] we can migrate jobs to data centers that are experiencing environments that require less cooling needs and thus lowering water usage by a sizable degree.
Source	M. Imran, M. Ibrahim, M. S. U. Din, M. A. U. Rehman, and B. S. Kim, "Live virtual machine migration: A survey, research challenges, and future directions," *Computers and Electrical Engineering*, vol. 103, p. 108297, 2022, doi: 10.1016/j.compeleceng.2022.108297.

Fairness view: Activity Diagram* of Job Processing



*Notations inspired by: [1]M. Seidl, M. Scholz, C. Huemer, and G. Kappel, '7 The Activity Diagram', in *UML @ Classroom: An Introduction to Object-Oriented Modeling*, 1st ed. 2015., in Undergraduate Topics in Computer Science. , Cham: Springer International Publishing : Imprint: Springer, 2015. doi: 10.1007/978-3-319-12742-2.

Fairness view (cont.)

Our proposed system uses active reporting and tracking of different data center parameters to calculate a desirability value for each data center depending on Alice's needs for her job. The desirability value is calculated by a function that takes into account the environmental parameters**(KPI 1)**, the data center's performance metrics and the current costs of using the data center in question**(KPI 4)**. It is a hot swappable function that can prioritize the wants of the user, where it can opt to favor environmental friendliness over all others factors or find the option that balances environmental concerns and performance over cost, etc.

To avoid overworking specific data centers or cause scheduling delays**(KPI 2)** the system also implements an active check for all machine learning jobs in progress. By continually checking job progress and if the data center still matches with Alice's needs, the system knows if it can keep the job running on the current data center or pause and reassign the job to a more fitting one**(KPI 3)**. The system also never has to opt out of a data center due to it approaching peak hours as it can reassign on the fly, assuring a fast scheduling time and start of any machine learning job. This ensures fairness in job assignment and data center selection by constantly re-assessing the situation.

Activity Diagram Explanation (Colours corresponding to AT-1, AT-2, AT-3, AT-4 as in the diagram)

- **Full job processing and assignment flow (Main flow):**
  - **A1:** The system received a training job and accepts it for analysis
  - **A2:** System checks job parameters, calculates expected run time is calculated and checks requests concerning DC parameters
  - **A3:** System sets the criteria for the job
  - **A4:** Scheduler receives job and prepares it for assignment
  - **A5:** Retrieves all known data centers
  - **A6 → D1 → A7:** The first in line data center is picked if it exists and the system requests its parameters
  - **A8, A9, A10:** Parallel to each other retrieves the environmental factors, performance data and costs
  - **A11:** Applies the desirability function and returns value
  - **A12 → D2 → A13:** Analyzes return value to see if it is within an acceptable range and assigns the job to the DC if so.
  - **A14:** Job is now in progress
  - **A15 → D3 → A16:** Checks job progress and terminates the job if completed.

- **Criteria too strict for any data center:**
  - **D1 → A17:** In case none of the data centers match requested criteria, we loosen some of the criteria less prioritized by the engineer
  - **A17 → M1 → A3:** The job with loosened criteria is sent back to the scheduler and merges with the main flow via M2

- **Data center does not match criteria:**
  - **D2 → A18:** System skips the data center and attempts to select the next data center on the list
  - **A18 → M3 → A6**: The path merges with the main flow via M3 and the next data center is selected

- **Job not finished during completion check, but near completion:**
  - **D3 → A19:** We check the progress of the job and compare it to the expected job duration
  - **A19 → D6 → A20:** If the job is nearing completion then the system should not waste resources trying to reassign the job to a different data center
  - **A29 → M4 → A14:** System lets the job run its course on the current data center and path returns to the main flow via M4

- **Job not finished and not near completion but data center is within parameters: (KPI 3)**
  - **D6 → A21**: System estimates job is not near completion
  - **A21 → D5 → A20:** Data center parameters are still acceptable
  - **A20 → M4 → A14:** Job continues and the path joins the main flow via M4

- **Job not finished and not near completion and data center is not within parameters: (KPI 3)**
  - **D6 → A21:** System estimates job is not near completion
  - **A21 → D5 →A22:** Data center parameters are not acceptable
  - **A22 → M1 → A3:** Job is paused and sent for rescheduling, path rejoins the main flow via M1

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